

Test-Retest Reliability of the Open Card Sorting Method

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ABSTRACT

Open card sorting is a popular method for designing information architectures. However, there is a lack of empirical evidence on the validity and reliability of the method. This paper explores the test-retest reliability of open card sorting. To this end, a within-subjects study was conducted. The same participants performed open card sorts twice with a time interval of 15-20 days. Content from three website domains was used: an eshop, a travel and tourism website, and a university website. Results showed that participants provided highly similar groupings and labels between the two card sorts per domain. A high agreement of the produced navigation structures was also found. These findings provide support for the test-retest reliability of open card sorting.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Empirical studies in HCI.

KEYWORDS

Card sorting, Information architecture, Test-retest reliability, Method evaluation

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1 INTRODUCTION

Card sorting is an important method in HCI research and practice for the design or evaluation of Information Architectures (IAs) [2, 26, 31, 32]. The IA of an interactive system describes the way its content is organized and labeled [32]. Back in 1985, Tullis [33] used card sorting to organize the menus of a mainframe operating system, so it is a method that has been around in HCI for more than 35 years. More recently, the method became popular for supporting the IA design of websites [11, 22, 29, 39]. A well-planned IA increases the usability of a website by reducing the time and effort required to find

information [29]. Card sorting has been also used to organize design guidelines [1, 13, 38], to validate tools for automated grouping of content items in websites [14, 15] and applications in multipurpose public displays [17], and to understand users' mental models in various contexts, such as web developers' programming knowledge [9], experts' perspectives on haptic devices [30] and self-tracking for mental wellness [18], and children's perceptions of cybersecurity warnings [12] and mobile games categorization [4].

Participants in a card sort are provided with content items (cards) and they have to group them in categories. There are two primary variations of card sorting: open and closed. In an open card sort, participants group the provided cards into their own categories, and they provide a label for each category they create. In closed card sorts, participants sort the provided cards into a set of predefined labeled categories. Other card sorting variations also exist. The hybrid variation [11] is a combination of open and closed card sorting in which users can add new categories to some predefined ones. In the Modified-Delphi card sorting variation [25], the first user conducts an open sort and each next one changes the sorting until consensus is built.

Open card sorting is the most widely-used variation [31, 37]. Various methodological issues related to open card sorts have been researched and are discussed in the following section. Surprisingly, there is a lack of empirical evidence on the method's validity and reliability. There is only one study [16] that supports the cross-study reliability of open card sorts by comparing the sortings on the same content completed by different samples of similar profile participants. However, more popular aspects of a method's reliability, such as consistency of results obtained from the same users across time, known as test-retest reliability or repeatability, have not been addressed. This paper investigates the test-retest reliability of open card sorts and the following research questions in specific:

- RQ1: Does the open card sorting method produce consistent data when it is repeated on the same sample of people at a different point in time?
- RQ2: Does analysis of open card sorting data produced by the same sample of people at different time points result in similar navigation structures?

2 RELATED WORK

Previous research has investigated various methodological issues related to open card sorts. Regarding the sample size, 15 to 20 users are enough to obtain reliable data [23, 35]. The number of cards to be sorted has been also investigated [31, 34]. Spencer [31] argues that it should be neither less than 30, because creating groups can be difficult, nor larger than 100, as it may be tiring or confusing for users. For a large number of cards, Tullis and Wood [34] suggest having each person sort only a 60% sample of the full set. They

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found that if 30-40 participants are involved in this sub-sample sorting then the obtained results are equivalent to sorting done on the full set of cards.

Regarding the amount of time required to perform a sort, Hudson [11] mentions that it is largely dependent on the number of items to be sorted: approximately 20 mins for 30 cards, 30 mins for 50 cards and 60 mins for 100 cards. Card sorting can be conducted either manually using physical cards or using software [28]. Software-mediated card sorting reduces the effort for data collection and the potential for data entry mistakes, and makes the recruitment process easier [5]. Studies [10, 27] have found no difference between the results obtained by manual and software-mediated card sorts, but the latter took significantly longer [27]. The usability of card sorting tools has been also investigated and different tools were preferred by researcher and user participants [5].

A rather large body of literature exists on analysis of open card sort data. Research has shown that analyzing the same data using different methods leads to different IAs [22]. Righi et al. [28] describe best practices for analyzing data from open card sorting exercises. Spencer [31] argues that the data can be analyzed using exploratory (qualitative) or statistical (quantitative) analysis or a combination of both. Exploratory analysis typically uses tabulations

of the data, such as items×items and items×categories matrixes, and helps to find patterns in the collected data that show how participants think about the cards and the groups. Paul [26] describes a method based on node-link graph visualizations to support in-depth qualitative analysis that is scalable and repeatable.

Various techniques are used for the statistical analysis of open card sort data, such as hierarchical cluster analysis [2, 8, 27, 31], k-means clustering [24, 31], multidimensional scaling [2, 24, 31], and factor analysis [3]. However, hierarchical cluster analysis is the most popular one [11, 22, 28, 37], and the average-linkage method is typically used because it produces balanced clusters that are easier to interpret [2, 36]. The main output of hierarchical cluster analysis is a dendrogram, a tree diagram that presents a hierarchy of groups based on content items' dissimilarity. This dendrogram can provide insights on organization of content and menu structures.

3 METHODOLOGY

A within-subjects study was conducted to compare the obtained results between two trials of card sort exercises. Cards from three general-purpose website domains were chosen and sorted twice by the same 16 participants. The collected data were analyzed and

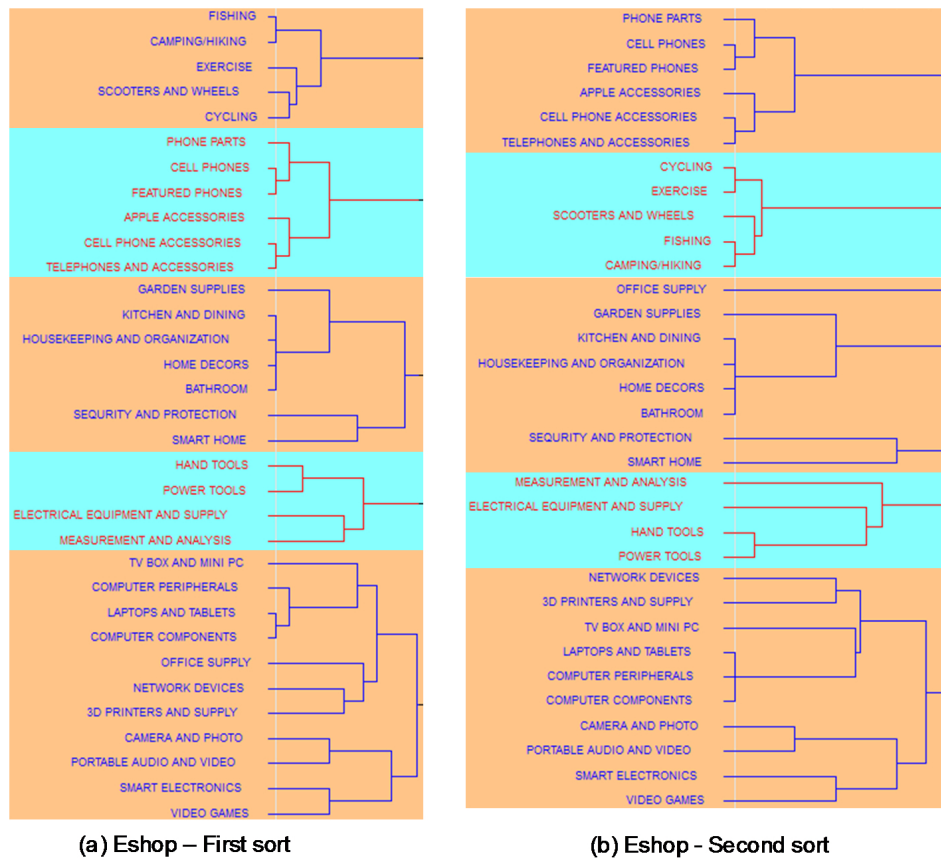


Figure 1: The average-linkage dendrograms produced for the first (left) and second (right) open card sort of the eshop. The dendrograms are cut in the number of categories that were selected for the proposed navigation structures.

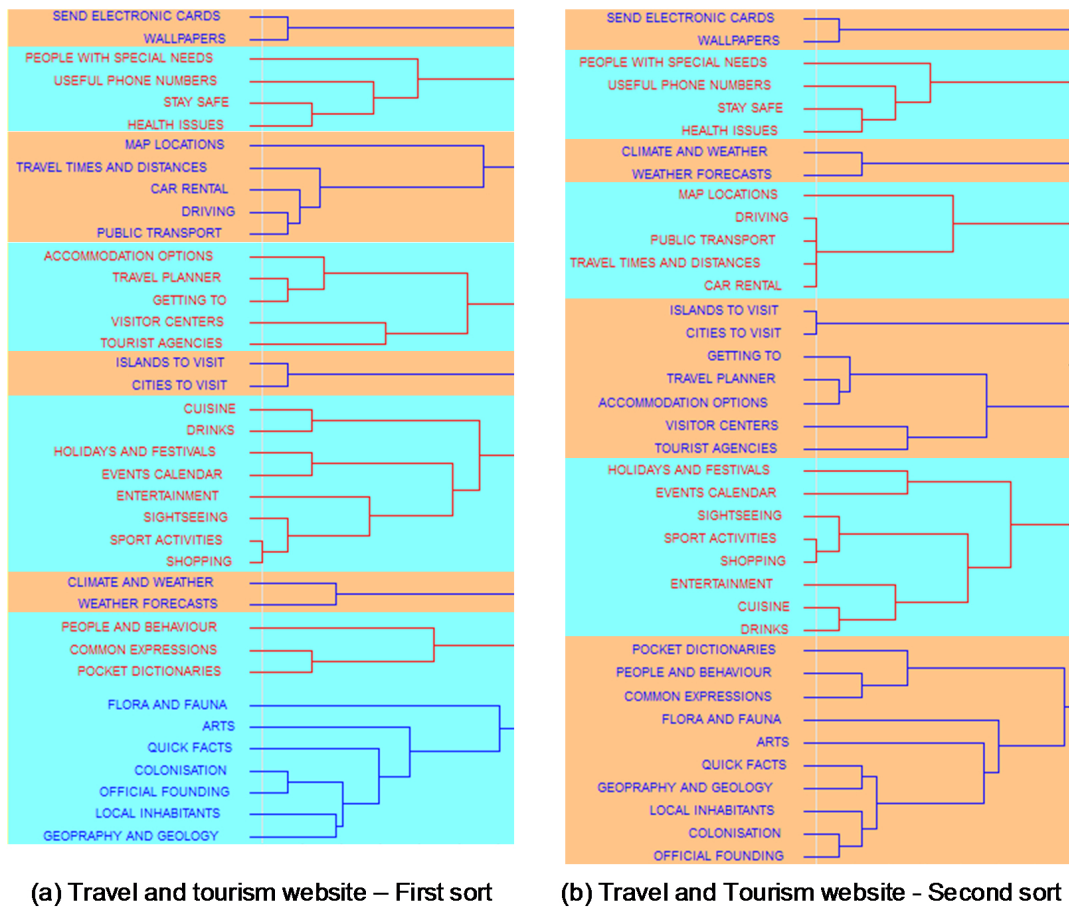


Figure 2: The average-linkage dendrograms produced for the first (left) and second (right) open card sort of the travel and tourism website. The dendrograms are cut in the number of categories that were selected for the proposed navigation structures.

compared using EZCalc [8], SPSS Statistics v25.0, Microsoft Excel 2016 and a Python implementation of the Mantel Test (<https://github.com/jwcarr/mantel>).

3.1 Participants

Sixteen users, 7 female and 9 male, aged from 20 to 55 years old ($M=28.6$, $SD=11.1$) participated in the card sorts. Research suggests a minimum of 15 users to obtain robust data from open card sorts [23, 35], thus our study had adequate sample size. All participants had good-to-excellent knowledge of English; the language used for the cards.

3.2 Website domains and cards

The first card sort was about an eshop and involved 33 cards, such as “Network devices” and “Computer peripherals”. The second card sort dealt with the design of a travel and tourism website. The participants were asked to sort 38 content items, such as “Islands to visit” and “Entertainment”. The third card sort was about a university website. The participants were asked to sort 28 content items, such as “Find a course” and “Campuses”.

3.3 Card sort data analysis

The collected card sorting data for each website were analyzed combining exploratory and statistical analysis according to the detailed guidance provided by Spencer [31]. First, analysis of the collected labels for the various groups proposed by the participants was conducted. To this end, a spreadsheet was created by examining the 16 sorts per condition one by one. In this spreadsheet, the rows represent the cards, the columns the group labels provided by one or more participants, and an intersecting cell the percentage by which a card appeared in the group with that label. For similar labels, a short list of all versions was created and the most used version was selected as the final label (standardized label). In the end, the most used labels for each group created were chosen.

Once the possible labels were chosen, the next step was to find the suggested groupings. We used EZCalc [8] to employ average-linkage hierarchical clustering to our card sort data and produce dendrograms. We also used SPSS to conduct factor analysis and find the statistically optimal, in terms of variance explained, number of clusters. Next, each dendrogram was first cut at the point in which

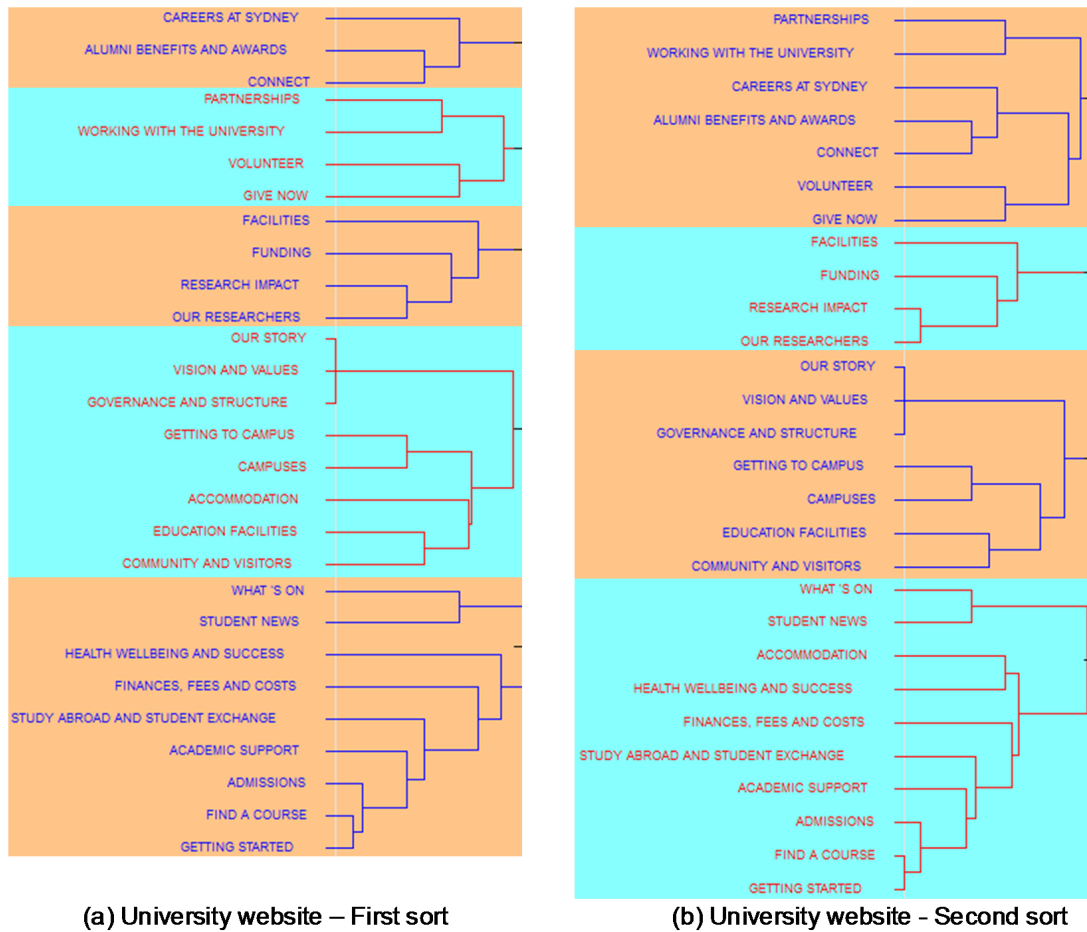


Figure 3: The average-linkage dendrograms produced for the first (left) and second (right) open card sort of the university website. The dendrograms are cut in the number of categories that were selected for the proposed navigation structures.

the statistically optimal number of groups appeared, and then an appropriate label from the ones produced in the exploratory analysis was selected for each group. If there were groups that no label matched with them, then the number of groups in the dendrogram was iteratively adjusted until an appropriate label could be found for these groups. This process resulted in two proposed navigation structures for each website, one per card sort exercise (Figure 1, Figure 2 and Figure 3).

4 RESULTS

The test-retest reliability of the open card sorting method was examined with two different types of comparisons: a) comparison of the raw open card sort data, and b) comparison of the navigation structures produced by performing state-of-the-art analysis of the open card sort data. Although these two comparisons are not orthogonal, they provide useful insights from different perspectives. The raw data comparison is a stringent objective approach, but the navigation structures comparison concerns the data-driven subjective output of the method.

4.1 RQ1: Open card sort data from the same sample of people at different points in time

Table 1 shows descriptive statistics of the groups created per condition. A Mann-Whitney test showed that the number of groups created by the participants did not differ significantly between the

Table 1: Descriptive statistics of the number of groups created per condition.

Website domain	Condition	M	SD	Mdn	Min	Max
Eshop	First sort	6.8	1.8	6.5	4	9
	Second sort	6.8	1.4	6.0	5	9
Travel and tourism	First sort	8.1	2.5	8.0	4	12
	Second sort	7.8	2.5	7.5	4	12
University	First sort	5.6	1.4	5.5	4	8
	Second sort	5.4	1.1	5.5	4	7

Table 2: Descriptive statistics of the labels' comparison between the two card sort trials per website domain.

Website domain	Identical or differ only in grammar	Synonyms	Supersets-subsets	Different
Eshop	M=70%, SD=20%	M=15%, SD=12%	M=6%, SD=8%	M=9%, SD=13%
Travel and tourism	M=67%, SD=19%	M=14%, SD=13%	M=7%, SD=10%	M=12%, SD=16%
University	M=76%, SD=23%	M=9%, SD=15%	M=2%, SD=5%	M=13%, SD=17%

two card sorts for all three websites; eshop: $z=0.000$, $p=1.000$, travel website: $z=1.089$, $p=0.276$, university website: $z=1.134$, $p=0.257$. A non-parametric test was used because the assumption of normality was violated in all cases.

We used the minimum Edit distance metric [7] to objectively compare users' groupings between the two card sort trials. This metric measures the distance between two sorts and is equal to the minimum number of cards that must be moved to turn the one sort into the other. Similarly to [27], we express this distance as a percentage of the total cards in the card set, which is easier to understand and unaffected by the different number of cards in the website domains. The mean minimum edit distance between the two card sort trials was 7.8% (SD=8.2%) for the eshop, 8.2% (SD=10.5%) for the travel and tourism website and 10.0% (SD=9.1%) for the university website.

Card sort data are typically summarized as a cards $\times$ cards distance or dissimilarity matrix [28]. Each cell value in this matrix measures the dissimilarity between two cards and ranges from 0 (cards were always grouped together) to 1 (cards were never grouped together). We used the Mantel test [20] to check the null

hypothesis that the distance matrices produced by the same users involved in the two card sort trials are not correlated. Unlike typical Pearson's or Spearman's correlation analysis, the Mantel test does not assume independence of observations or homogeneity of variance, which are very likely violated for a distance matrix [16, 27]. In our dataset, Mantel tests using 10000 permutations found a significant and very strong [6] correlation between the two distance matrices of each website; eshop: $r=0.98$, $p<0.001$, travel website: $r=0.96$, $p<0.001$, university website: $r=0.95$, $p<0.001$.

Finally, we compared the labels produced by the participants in the two card sort trials. For this comparison, we counted the number of labels that were: a) identical or differed only in grammar (e.g., "Phone" vs. "Phones", "Keep in touch" vs. "Keeping in touch"), b) synonyms (e.g., "Activity equipment" vs. "Activity gear", "Places to visit" vs. "Places to go"), c) supersets-subsets (e.g., "Cards" vs. "Cards and wallpapers", "Tools" vs. "Tools and supply"), and d) different. Next, we expressed these counts as a percentage of the total labels created per participant, which is easier to understand and unaffected by the different number of labels produced by the participants. Table 2 presents the obtained results per website domain. In sum,

Table 3: The two proposed navigation structures per website based on participants' card sort trials. Curly brackets represent the groups and each number represents a specific card. Differences in the two structures are presented in bold and underlined format.

Website domain	First card sort groups	Second card sort groups	Minimum number of cards to be moved
Eshop	A1 = {29, 30, 31, 32, 33} A2 = {1, 2, 3, 4, 5, 6} A3 = {16, 18, 24, 25, 26, 27, 28} A4 = {20, 21, 22, 23} A5 = {7, 8, 9 , 10, 11, 12, 13, 14, 15, 17, 19}	B1 = {29, 30, 31, 32, 33} B2 = {1, 2, 3, 4, 5, 6} B3 = { 9 , 16, 18, 24, 25, 26, 27, 28} B4 = {20, 21, 22, 23} B5 = {7, 8, 10, 11, 12, 13, 14, 15, 17, 19}	Card 9 from A5 to A3
Travel and tourism	A1 = {32, 33} A2 = {19, 20, 22, 36} A3 = {12, 18, 27, 29, 37} A4 = {1, 17, 23, 26, 38} A5 = { 8 , 10 } A6 = {6, 7, 15, 21, 24, 25, 30, 31} A7 = {13, 34} A8 = {2, 3, 4, 5, 9, 11, 14, 16, 28, 35}	B1 = {32, 33} B2 = {19, 20, 22, 36} B3 = {12, 18, 27, 29, 37} B4 = {1, 8 , 10 , 17, 23, 26, 38} B5 = {} B6 = {6, 7, 15, 21, 24, 25, 30, 31} B7 = {13, 34} B5 = {2, 3, 4, 5, 9, 11, 14, 16, 28, 35}	Card 8 from A5 to A4 Card 10 from A5 to A4
University	A1 = { 19 , 21 , 23 } A2 = {17, 20, 22, 24} A3 = {25, 26, 27, 28} A4 = {7, 8 , 11, 13, 14, 15, 16, 18} A5 = {1, 2, 3, 4, 5, 6, 9, 10, 12}	B1 = {} B2 = {17, 19 , 20, 21 , 22, 23 , 24} B3 = {25, 26, 27, 28} B4 = {7, 11, 13, 14, 15, 16, 18} B5 = {1, 2, 3, 4, 5, 6, 8 , 9, 10, 12}	Card 19 from A1 to A2 Card 21 from A1 to A2 Card 23 from A1 to A2 Card 8 from A4 to A5

the labels' similarity (those classified as identical or synonyms) between the two card sort trials was 85% (SD=16%) for the eshop, 81% (SD=18%) for the travel and tourism website and 85% (SD=18%) for the university website.

These findings support that the card sorting method produces consistent content groupings and category labels when it is repeated on the same sample at a different point in time.

4.2 RQ2: Navigation structures from open card sorts involving the same sample of people at different points in time

Figure 1, Figure 2 and Figure 3 show the average-linkage dendrograms cut at the number of categories that were selected for the proposed navigation structure per card sort of each website domain. From a visual inspection of these figures, one may support that the two navigation structures proposed per website seem highly similar.

We again used the minimum Edit distance metric [7], this time to objectively compare the two proposed navigation structures for each website. Table 3 presents the calculation steps for the minimum number of moves required to turn the navigation structure of the first sort into the one of the second sort, but the same result would be found for the vice versa conversion. In the case of the eshop, the card 9 (Office supply) must be moved from group A5 (Electronics) to group A3 (Home), resulting in an edit distance of 3.0% (1/33). For the tourism and travel website, the cards 8 (Islands to visit) and 10 (Cities to visit) need to be moved from group A5 (Places to visit) to group A4 (Planning the trip), thus the edit distance is 5.3% (2/38). Regarding the university website, the cards 19 (Careers at Sydney), 21 (Alumni benefits and awards) and 23 (Connect) must be moved from group A1 (Alumni) to group A2 (Get involved). In addition, the card 8 (Accommodation) needs to be moved from group A4 (About the university) to group A5 (Students). Thus, the edit distance for the university website is equal to 14.3% (4/28).

These findings provide evidence that the two navigation structures produced for each website by analyzing card sort data generated by the same participants at different time points were rather similar.

5 CONCLUSIONS

The purpose of this paper was to evaluate the test-retest reliability of the open card sorting method. The participants of this study completed three card sorts, one per website domain examined (eshop, travel and tourism, university), and repeated the same three

sorts 15-20 days later. Next, participants' raw data and the final navigation structures produced by the two card sorts per domain were compared to measure the test-retest reliability of the method.

Table 4 summarizes the findings of this study. Regarding the raw data comparison, participants' content groupings and labels were highly similar; at least 90% and 81% respectively. In addition, Mantel tests showed that the open card sort distance matrices were significantly ($p < 0.0001$) and highly (r at least 0.95) associated for all three domains. Furthermore, the navigation structures produced by performing state-of-the-art analysis of the card sort data were also highly similar (at least 86%). In specific, a small number of cards (from 1 to 4) had to be moved so that the two navigation structures per website domain would be identical.

In conclusion, this study found that the two card sort trials performed by the same participants at a time interval of 15-20 days were very similar to each other. The latter tends to support that the open card sorting method has high test-retest reliability. This finding is quite valuable because it provides scientific support for one aspect of the reliability of a widely used method that has been around in HCI for more than 35 years [33].

This study is part of our larger research efforts to provide empirical evidence on the validity and reliability of the open card sorting method, one valuable HCI method for IA design and evaluation. We have developed our own web-based card sorting tool, named CardSorter [21], to facilitate our data collection efforts. CardSorter is provided free of charge (<http://usability.csd.auth.gr/cardsorter>) and is open source (<https://github.com/CardSorter>).

Future directions of this research are to investigate the test-retest reliability of open card sorts with a rather large number of cards (e.g., more than 80) or for specialized topics (e.g., medical issues). In addition, we are currently running an ongoing study that investigates the validity of the method. To this end, we have selected five websites with IA-related usability issues and involved 30 participants in a user testing experiment collecting traditional usability measures. We plan to engage a different second sample of participants to perform open card sorts for these websites and produce a redesigned IA based on card sort data analysis. Then, we will involve the first sample of the same 30 participants in a user testing experiment of the redesigned websites and compare the collected usability measures. If a significant increase is observed in these measures, then it can be attributed to the IA redesign based on the open card sort, which in turn would provide support that the method is valid for improving the usability of IAs. Finally, we also plan to investigate the effect of participants' personality traits [19] on the method's effectiveness to produce usable IAs.

Table 4: The findings of this within-subjects study that investigated the test-retest reliability of the open card sorting method. The same participants sorted the content items of three websites at two points in time.

Website domain	Mean minimum edit distance of participants' sorts	Distance matrices correlation	Mean similarity of participants' labels	Minimum edit distance of navigation structures
Eshop	7.8% (SD=8.2%)	0.98 ($p < 0.001$)	85% (SD=16%)	3.0%
Travel and tourism	8.2% (SD=10.5%)	0.96 ($p < 0.001$)	81% (SD=18%)	5.3%
University	10.0% (SD=9.1%)	0.95 ($p < 0.001$)	85% (SD=18%)	14.3%

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